

Lab Journal Exercises

Consists of exercises from Units Processes, Modelling and Data

Group 6

ENVM1502 Project River Basin Hydrology and Water Management

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Summary of Exercises

Exercise Unit	Name	File name
1 Processes	Catchment Functioning, Macroscale Laws & Change	Exercise_Processes_1&2(...)
2 Processes	Vegetation Processes & Change	Exercise_Processes_1&2(...)
3 Data	Watershed & Stream Delineation	-
4 Processes	Precipitation and Flood Risk	Exercise_Processes_3(...)
5 Processes	Cold Region Processes and Change	Exercise_Processes_4(...)
6 Modelling	eWaterCycle	EWaterCycle1-5
7 Data	Likelihood-Based Model Calibration Data	Exercise_Data_Likelihoodbased
8 Data	Closing Water Balance Using Remote Sensing Data	Exercise_Data_Closingthewaterbalance
9 Modelling	FlexTopo	Exercise_Flextopo (separate folder)
10 Data	Land Use Classification	-

Data sources

- **CHIRPS Daily Precipitation**

Climate Hazards Group. (n.d.). *CHIRPS: Climate Hazards Group InfraRed Precipitation with Station data*. University of California, Santa Barbara. <https://www.chc.ucsb.edu/data/chirps>

- **ERA5 Climate Reanalysis (Precipitation & Evaporation)**

Hersbach, H., Bell, B., Berrisford, P., et al. (2020). *ERA5 hourly data on single levels from 1959 to present*. Copernicus Climate Change Service (C3S). <https://doi.org/10.24381/cds.adbb2d47>

- **GLDAS (Global Land Data Assimilation System)**

Rodell, M., Houser, P. R., Jambor, U., Gottschalck, J., Mitchell, K., Meng, C. J., ... & Toll, D. (2004). *The Global Land Data Assimilation System*. Bulletin of the American Meteorological Society, **85**(3), 381–394. <https://ldas.gsfc.nasa.gov/gldas>

- **GRACE Terrestrial Water Storage**

NASA Jet Propulsion Laboratory. (n.d.). *GRACE Tellus Monthly Mass Grids*. <https://grace.jpl.nasa.gov>

- **MODIS Evapotranspiration (MOD16)**

Mu, Q., Zhao, M., & Running, S. W. (2011). *MODIS global evapotranspiration (ET) product (MOD16A2) NASA Earth Observing System, Version 6*. NASA EOSDIS Land Processes DAAC. <https://doi.org/10.5067/MODIS/MOD16A2.006>

- **IMERG Precipitation Data**

NASA. (n.d.). *IMERG: Integrated Multi-satellite Retrievals for GPM*. <https://gpm.nasa.gov/data/imerg>

- **HILDA+ Land Use Dataset**

CEOS. (n.d.). *HILDA+ Global Land Use Change Dataset*. Group on Earth Observations Global Stocktake. <https://ceos.org/gst/HILDAplus.html>

- **Observed Discharge**

Republic of the Philippines, Department of Public Works and Highways – Bureau of Design – Water Projects Division. (n.d.). *Streamflow Monitoring Stations*. https://apps.dpwh.gov.ph/streams_public/station_public.aspx

01 Catchment Functioning, Macroscale Laws & Change - Processes

Data used in this exercise

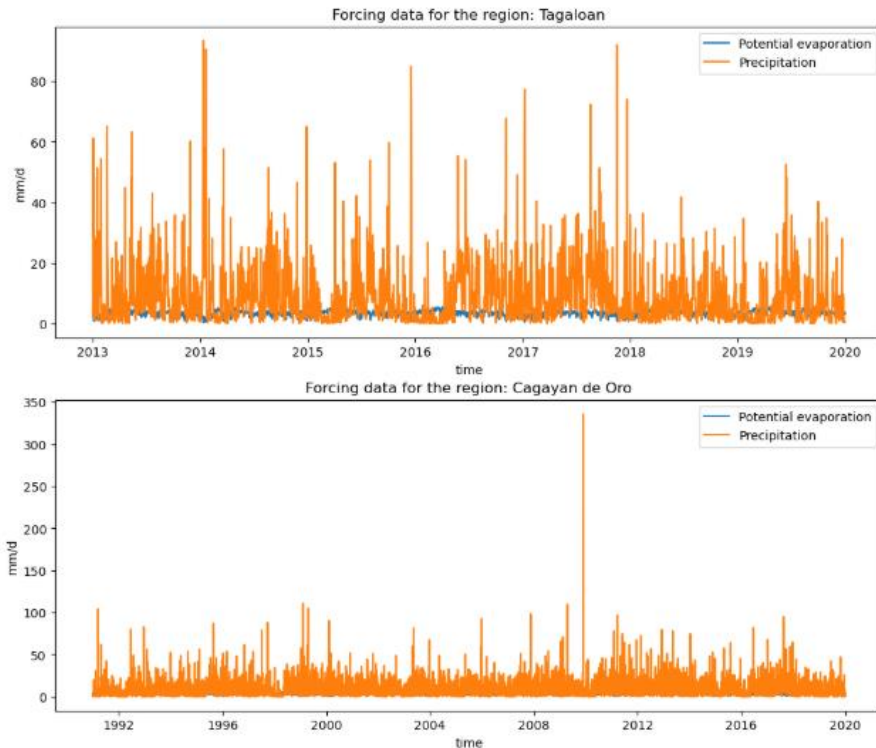
Cagayan de Oro:

- **Forcing data period:** 1991–2019
- **Data type:** precipitation, evaporation (ERA5), discharge (Philippine Government)
- **Analysis period (used for discharge comparison):** 2003–2012

Tagoloan:

- **Forcing data period:** 2013–2019
- **Data type:** precipitation, evaporation (ERA5), discharge (Philippine Government)
- **Analysis period (used for discharge comparison):** 2013–2019

We analyzed ERA5 forcing and discharge data for two catchments; Cagayan de Oro and Tagoloan to to characterize catchment response, evaluate water balance closure and compare evaporative behavior using the Budyko framework, flow duration curves (and autocorrelation functions).



“Analysis of daily Precipitation and Potential Evaporation from ERA5 for Tagoloan and Cagayan de Oro.”

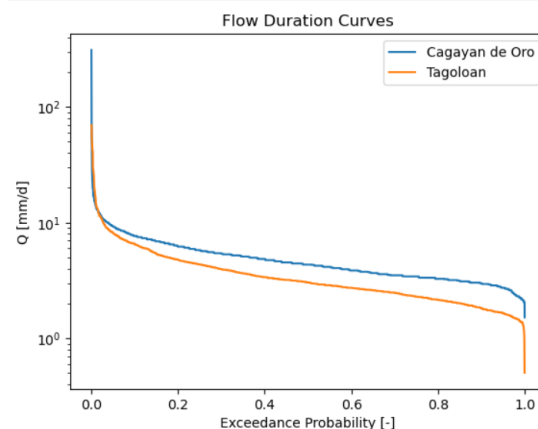
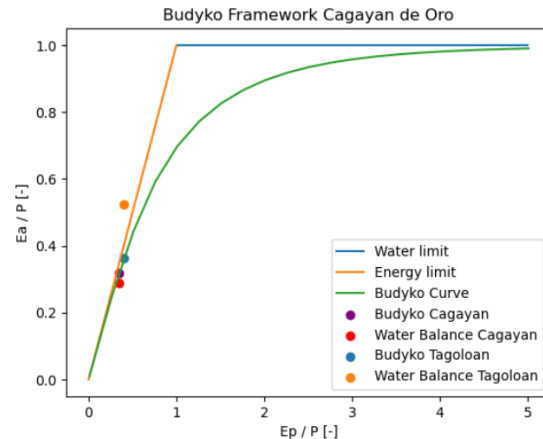
The figure shows daily time series of ERA5 forcing data, precipitation and potential evaporation, for the two catchments; Tagoloan and Cagayan de Oro. In both catchments, precipitation is highly variable with clear episodic peaks, while potential evaporation remains relatively low and stable, as expected in humid tropical climates.

In the Cagayan de Oro series, we observe a large precipitation peak around **24 November 2009**, exceeding 300 mm/day. Initially, we considered this could be due to a tropical storm event; however, after some research, no records of such an event were found for that specific date. Moreover, if this extreme rainfall were real, a corresponding increase in discharge would be expected shortly afterward. We looked into that, but no such response is seen in the discharge data.

This inconsistency strongly suggests that the observed peak is likely a data error, possibly from sensor malfunction or incorrect reanalysis interpolation.

01 Catchment Functioning, Macroscale Laws & Change - Processes

“This figure shows the Budyko framework, which relates long-term water and energy availability in the catchments.”



Each region is represented with two points:

- The Budyko estimate (purple for Cagayan)
→ calculated using only P and E_p and assuming Budyko-type balance.
- The observed water balance point (red for Cagayan)
→ derived from observed discharge Q via $E_a = P - Q$

Interpretation of the graph

Both points fall within the water-limited region ($E_p/P > 1$), consistent with a tropical catchment experiencing a seasonal dry period. The Budyko-estimated point lies exactly on the curve, validating the expected theoretical behavior. However, the observed water balance point falls slightly below the curve, suggesting that either actual evaporation is higher than expected or observed discharge underestimates, potentially due to measurement bias, model simplification, or unaccounted catchment losses such as deep percolation or anthropogenic use.

As can be seen the Tagoloan water balance plots outside of the Budyko framework, this can have one or more of the following reasons:

- Incorrect data
- Other in and outflows that are not accounted for
- A difference in the topographic and phreatic device

Conclusion:

The Budyko analysis confirms that the Cagayan de Oro basin operates in a water-limited regime, with evapotranspiration constrained by rainfall. The close alignment with the Budyko curve supports theoretical assumptions, but the slight deviation of the observed point hints at possible data limitations or additional processes not captured by the Budyko framework alone.

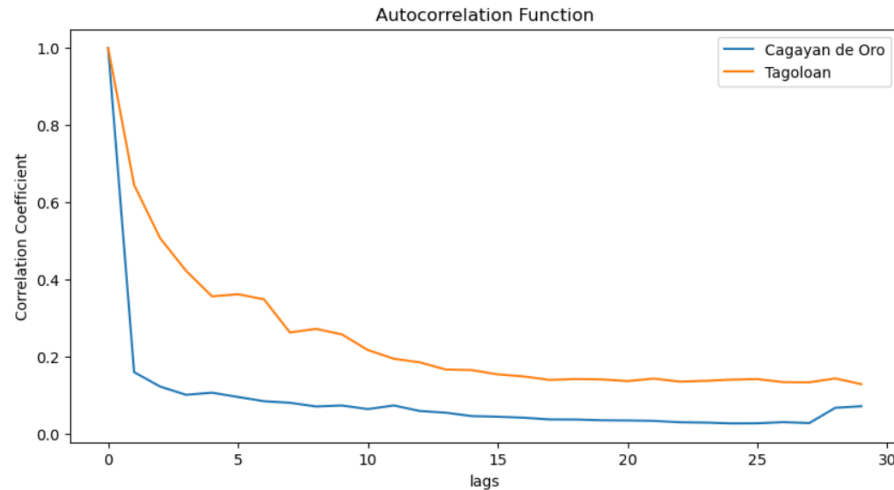
The second graph shows the Flow Duration Curves for the Cagayan de Oro and Tagoloan catchments.

Interpretation of the graph

- Cagayan de Oro (blue line) has consistently higher flows than Tagoloan (orange line) across almost the entire exceedance range, indicating a generally wetter or more responsive catchment.
- The steeper slope of the FDC for Tagoloan suggests greater flow variability, whereas the flatter curve of Cagayan implies a more stable flow regime.

01 Catchment Functioning, Macroscale Laws & Change - Processes

“This graph shows the autocorrelation function (ACF) of daily streamflow for the catchments Cagayan de Oro (blue) and Tagoloan (orange), plotted over a lag range of 0 to 30 days.”



Interpretation of the graph;

- Tagoloan shows a much slower decline in autocorrelation with increasing lag, staying above 0.2 even after 25 days.
→ This suggests more persistent discharge, possibly due to groundwater contributions, higher catchment storage or slower hydrological response.
- Cagayan de Oro shows a rapid drop in autocorrelation after just 1–2 days, levelling off below 0.1.
→ This indicates a flashier system with quick runoff generation and shorter memory, likely dominated by surface runoff and steeper topography.

Tagoloan shows greater flow persistence, suggesting it responds more slowly to rainfall and may be more resilient to dry spells. Cagayan de Oro responds more quickly but loses that memory fast, likely reflecting shorter flow paths and lower subsurface storage. These differences are important when comparing flood risks, water availability, and hydrological modelling approaches for the two basins.

02 Vegetation Processes & Change - Processes

Data used in this exercise

Cagayan de Oro:

- **Forcing data period:** 1991–1994 (pre deforestation) and 1995-1999 (post deforestation)
- **Data type:** Data type: precipitation, evaporation (ERA5), discharge (Philippine Government)

Results

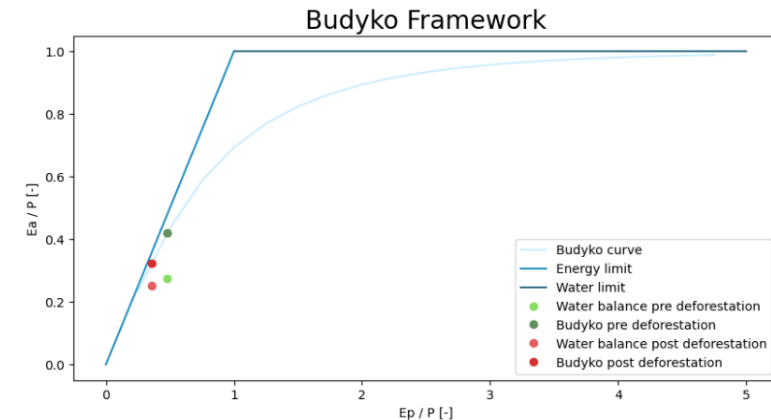
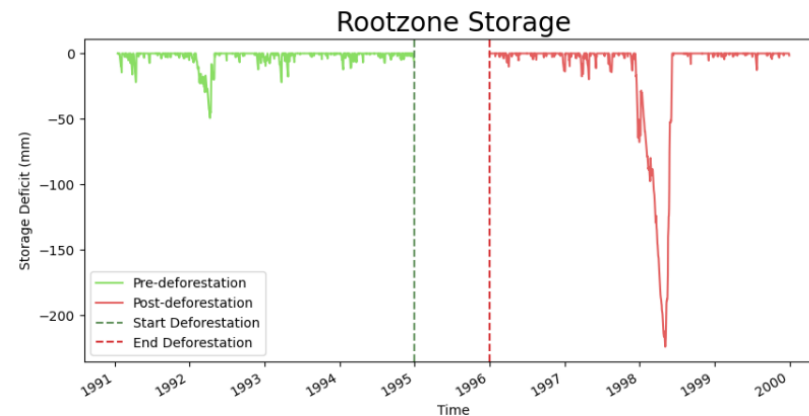
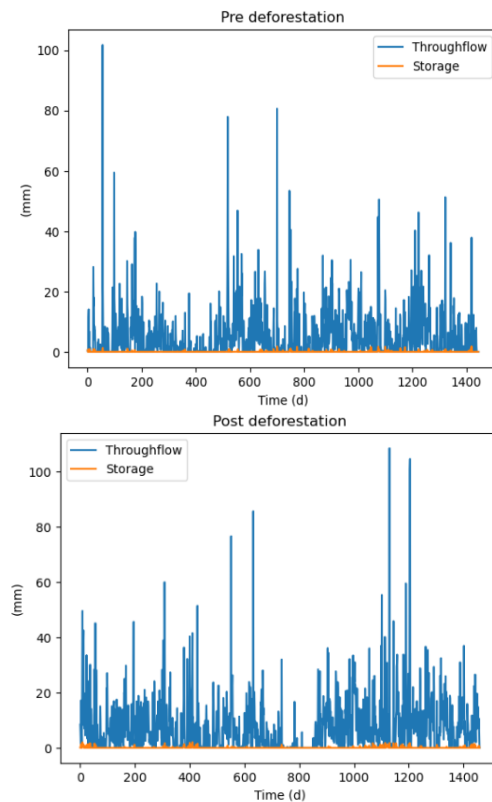
	Mean Pe / P (mm)	Sum Ei (mm)	Sum P (mm)	Sum S (mm)	Max Sr (mm)
Pre deforestation	0.73	2909.65	10585.74	64.92	170.97
Post deforestation	0.79	3894.46	18206	126.22	658.73

Analysis

We would expect to see a drop in storage (S) and rootzone storage (Sr). However, the opposite seems to be happening.

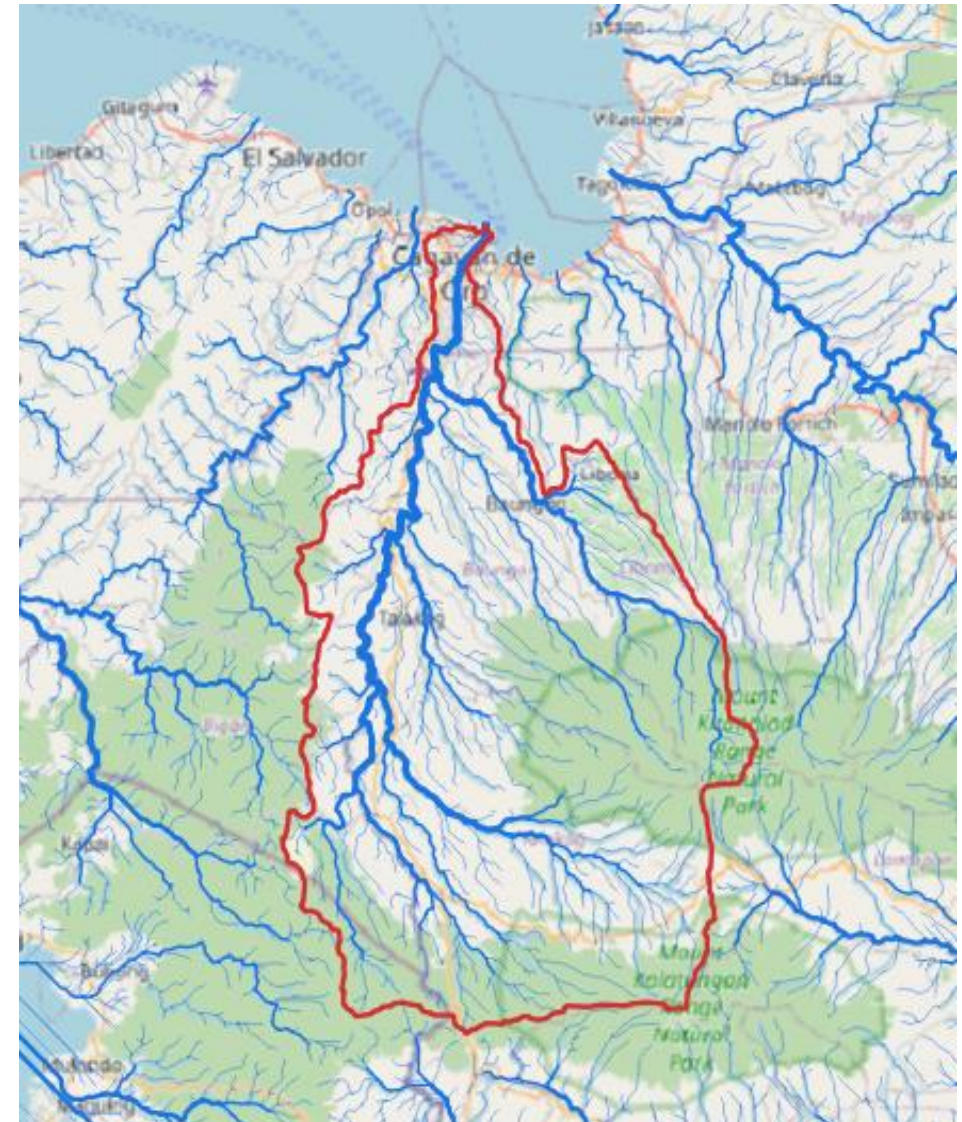
The increase in throughfall (Pe / P) does comply with what we expect would happen after deforestation. Because there are less trees to intercept the precipitation more reaches the soil.

In the Budyko framework we see a shift down after deforestation, because less interception also means less actual evaporation. We also see that the loss (Ea – Ep) becomes bigger after deforestation.



03 Watershed & Stream Delineation - Data

The Cagayan de Oro River Catchment is located in Northern Mindanao, draining toward the northern coast of the island into Macajalar Bay, part of the Bohol Sea. The river originates in the Kitanglad Mountain Range in the province of Bukidnon, and flows in a north-northwest direction through steep uplands, mid-elevation agricultural landscapes, and the densely populated city of Cagayan de Oro, before reaching the coast in Misamis Oriental. The catchment lies approximately between 8.1°N to 8.4°N latitude and 124.3°E to 124.7°E longitude.



04 Precipitation and Flood Risk - Processes

Data used in this exercise

- **Source:** ERA5 daily precipitation retrieved via eWaterCycle
- using LumpedMakkinkForcing for Cagayan de Oro.
- **Period:** 2003-01-01 to 2019-12-31
- **Resolution:** Daily
- **Annual Maxima Series** : One maximum rainfall value per year

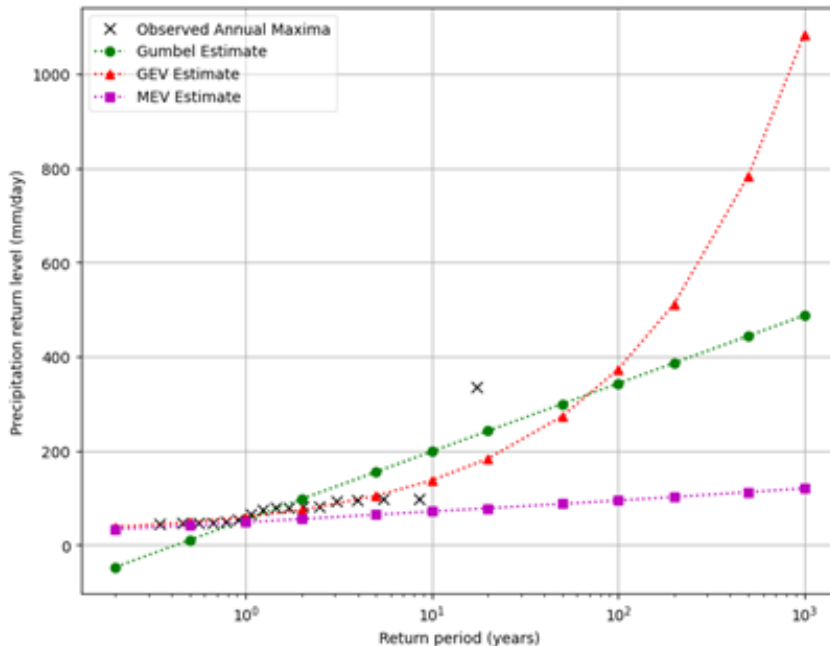
We used three methods for EVA in this exercise

- Gumbel distribution
- Generalized Extreme Value distribution (GEV)
- Metastatistical Extreme Value method (MEV)

We compare and interpret the behavior of the Gumbel, GEV and MEV extreme value methods in estimating precipitation extremes and associated uncertainties for our region. The analysis is based on data short observational record (17 years). This limits the number of extremes observed and thus has significant implications for return level estimation.

Return Level Plot – Comparing Gumbel, GEV, MEV

This plot compares the return level estimates from the three EVA methods.



- **Observed annual maxima**, which show the empirical block maxima. The standout point (~336 mm/day in 2009) is clearly visible as a statistical outlier. This is very likely associated with an intense tropical storm, which introduces a heavy-tailed behavior in the data.

1. Gumbel Estimate, this assumes a fixed distribution shape (no shape parameter).

- Provides a moderate extrapolation beyond observed data, but the distribution may underestimate rare extremes.
- This works well for small sample sizes but lacks flexibility in the tails.

2. GEV Estimate

- Incorporates a shape parameter to model heavy tails, allowing for more flexibility.
- Reacts very strongly to the 2009 extreme event, leading to very high estimates for long return periods (>100 years).
- This steep curve suggests that GEV may overestimate risk if the extreme is a one-off or measurement error. This is a known limitation of GEV with small datasets and influential outliers.

3. MEV Estimate, it shows the least steep growth in return level estimates, reflecting its robustness to the outlier and sampling variability.

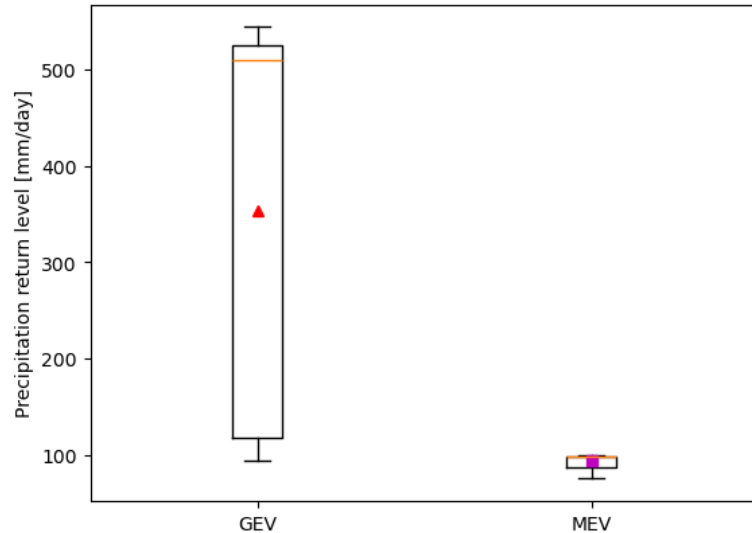
- The MEV curve suggests that unless many years with similarly extreme events occur, the 2009 event is not representative of the general climatology.
- This approach may be more conservative and appropriate for design decisions for our data-scarce contexts.

04 Precipitation and Flood Risk - Processes

Monte Carlo Boxplot – Uncertainty at 100-year return period

This following plot shows the spread in the 100-year return level estimate obtained from 500 Monte Carlo resampling's of only 10 random years each, which simulates short data availability.

Boxplot of the 100-year return period using 10 random years repeated 500 times



- **GEV** shows very high variability in its estimates; the box stretches from ~100 mm/day to >500 mm/day. This illustrates how unstable GEV can be under limited data, especially when one extreme event is randomly included (as in our case).
- **MEV** shows a narrow box, indicating a low spread in estimates. This stability arises from the method's use of all wet days per year and its lesser dependence on the one isolated extreme.
- The median estimate of MEV (+90 mm/day) is much lower than that of GEV, highlighting a conservative bias.

Finalizing all, this exercise analysis shows that:

- **GEV** provides flexibility but is sensitive to the outlier and sample size.
- **MEV** offers stability and leverages more data per year, making it well-suited to small-sample hydrology.

In the context of Cagayan de Oro, where flood risk is severe and data is limited, MEV may be more appropriate for conservative planning, while GEV may offer insight into worst-case scenarios. Using both methods together provides a more complete picture of flood hazard.

The analysis of this exercise provides us with valuable insight into the statistical behavior of extreme precipitation, but we did not directly use these results to answer our key research question regarding the impact of deforestation on peak discharge. The reason is that EVA focuses on the statistical characterization of precipitation extremes at the outlet and does not explicitly account for catchment-scale hydrological processes, such as runoff generation, infiltration and changes in peak discharge due to vegetation cover. However, it helps establish the meteorological boundary conditions under which flooding might occur and provides a quantitative basis for selecting extreme rainfall scenarios in hydrological modelling.

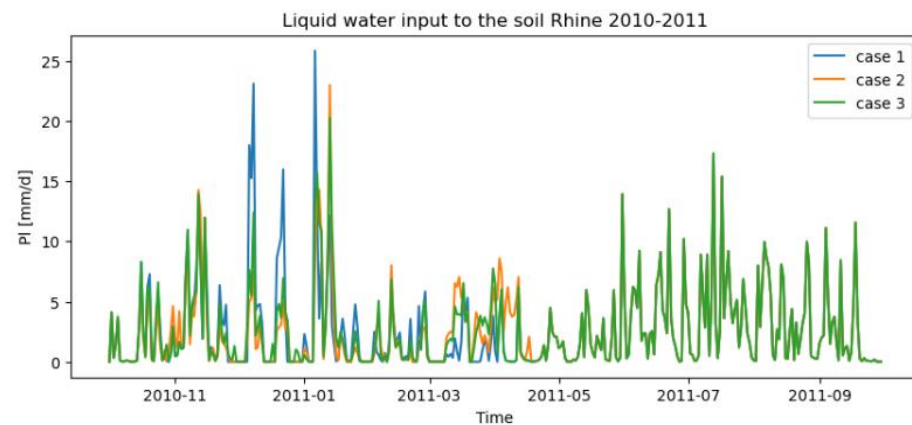
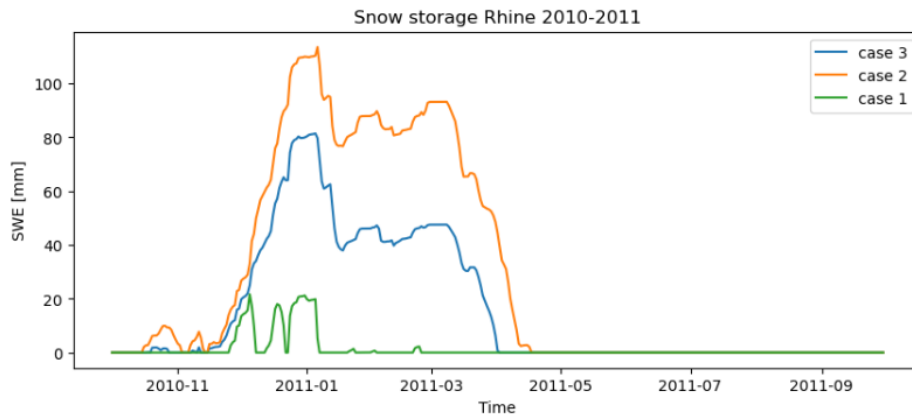
05 Cold Region Processes and Change - Processes

Data used in this exercise

Rhine:

- **Forcing data period:** 2010/10/01 – 2011/09/31
- **Data type:** temperature and precipitation (ERA5)

Results



Analysis

Case 1: flat terrain at 700m elevation.

Case 2: 4 elevation bands (0-500m, 500-1000m, 1000-1500m, 1500-2000m), 25% of basin area (**163122,5 km²**) per elevation band.

Case 3: similar to case 2 but with a 2-degrees temperature increase.

The mountainous catchment accumulates more snow as the temperature is lower at higher elevation, meaning more snow falls and snowmelt sets in much later. In the flat terrain catchment, very little snow falls and accumulates suggesting that temperatures never go down very far and most of the snow already melts during the winter.

Snowmelt in the flat terrain occurs at a higher rate as once the temperature is above 0 Celsius the snow in the entire catchment will start to melt. This then also results in a higher flood risk as the soil has less time to infiltrate all the water. However, this can be seen in the PI graph, where high peaks occur right when the snow melts in the SWE graph.

Comparing case 2 and 3, it can be seen that a temperature increase results in a decrease of the snow accumulation. The graph also becomes a bit less peaked (as the precipitation fluctuations do not necessarily translate to snowfall).

Notes

As snowfall was not relevant for our region in the Philippines, where temperatures did not drop below 0 degrees to allow for snowfall, we completed this exercise on data of the Rhine catchment.

Furthermore, for the second case the elevation bands defined in the exercise were used. As this exercise was not relevant for our research, we did not spend time to analyse how the elevation bands and area are distributed for this specific catchments. We understand this is not representative for this region, and therefore the results are not complete. However, it still enabled us to analyse the effects of the different elevations compared to a flat catchment.

06 eWaterCycle - Modelling

eWaterCycle platform was used to implement, run and calibrate a custom HBV rainfall-runoff model for the Cagayan de Oro catchment. The process consisted of several steps that mirror the hydrological modelling cycle: model setup, forcing data preparation, simulation, calibration and evaluation.

1. Model implementation; we created our own HBV model, following the BMI interface structure required by eWaterCycle.
2. ERA5 forcing generation; using eWaterCycle's integrated link with ESMValTool, we generated lumped ERA5 forcing data (precipitation and potential evaporation) for our study area. The input catchment boundary was provided through a shapefile.
3. Model run & calibration; the HBV model was run with ERA5 forcing and observed discharge data. We used this setup to evaluate model performance. Calibration was performed using a simple parameter sweep, evaluating using two objective functions: NSE and NSE. However, low NSE values and similarly poor log(NSE) scores indicated limited model performance, likely due to forcing data uncertainties, lack of soil moisture or land use representation or simplified model structure.
4. Comparison and interpretation; while the HBV model was structurally capable of simulating discharge, the poor NSE values revealed the challenges of capturing the catchment's fast hydrological response. The model performance suggests that further parameter tuning, refined forcing data or a more complex model may be necessary to accurately simulate hydrological dynamics in this tropical mountainous basin.
5. CMIP6 forcing generation for climate impact assessment

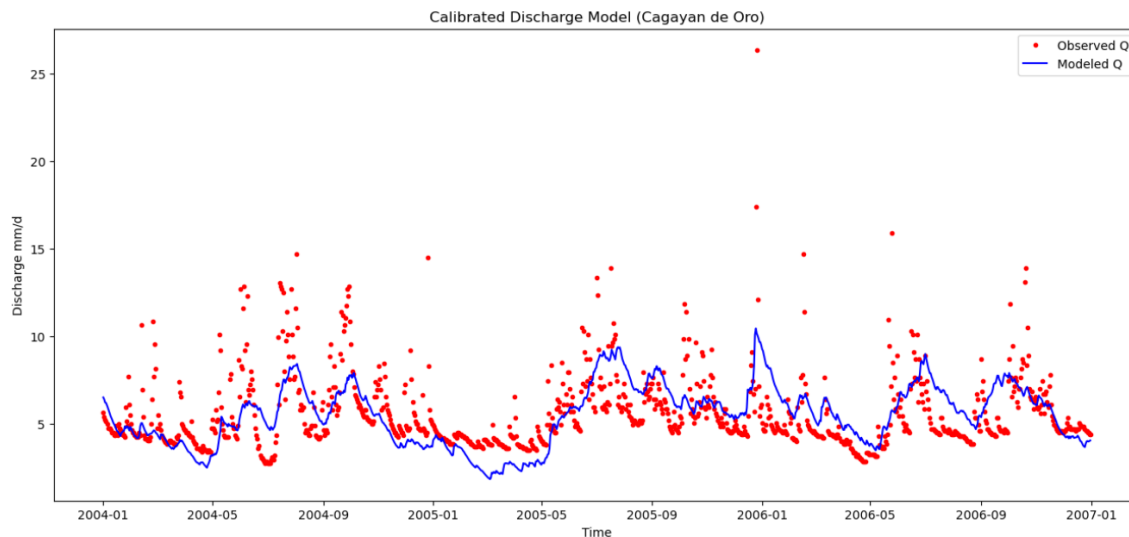
07 Likelihood-Based Model Calibration Data - Data

Data used in this exercise

Cagayan de Oro:

- **Data type:** precipitation, evaporation (ERA5), discharge 1991-2019 (Philippine Government)
- **QGIS files:** Shapefile of the Cagayan de Oro River Basin

Model Calibration Using NSE



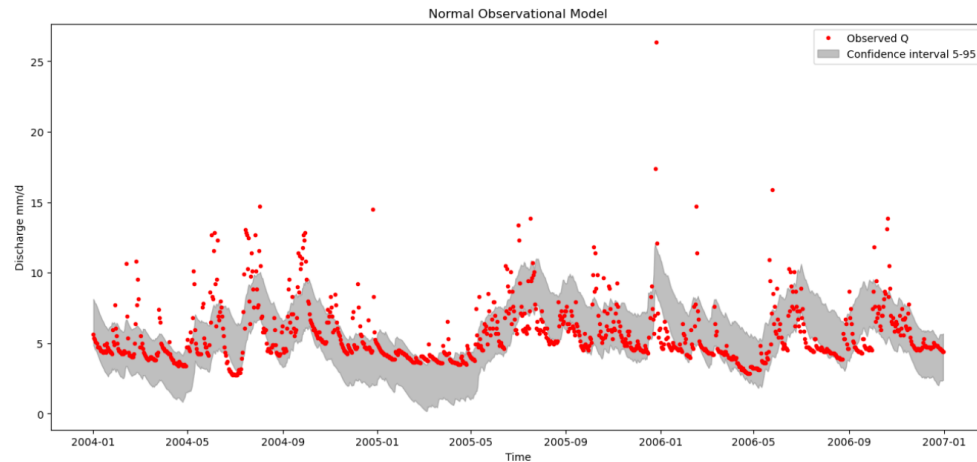
To evaluate the performance of the HBV model for the Cagayan de Oro River Basin, we performed calibration using the NSE as an objective function. After defining the function `neg_NSE()` and excluding the first year of simulation as a warm-up period, we used the differential evolution algorithm to identify the optimal model parameters. The resulting NSE was approximately 0.18, which indicates a **low to moderate model skill**. It's **not very sufficient** for operational forecasting or precise hydrological predictions.

In the time series plot covering 2004 to 2006, the **observed discharge** shows high variability, with pronounced peaks during the wet seasons. The **modeled discharge** is smoother, tracking the seasonal patterns fairly well but clearly underestimating several peak events.

This discrepancy could be due to the lumped structure of the HBV model, simplifications in evapotranspiration input or limitations in the rainfall data resolution and catchment representation.

07 Likelihood-Based Model Calibration Data - Data

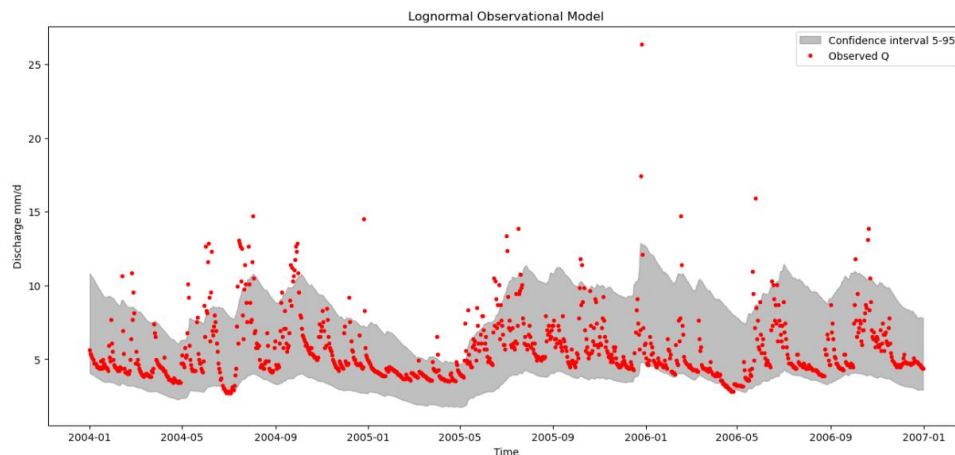
Likelihood-based Calibration



Here the model calibration is performed using a normal likelihood function, assuming residuals between observed and simulated discharge follow a Gaussian distribution. The grey band represents the 5–95% confidence interval of modeled discharge. The observed data (red dots) mostly fall within this interval, indicating the model captures the general discharge dynamics and uncertainty range reasonably well. However, several high-flow events exceed the upper bound, suggesting the normal assumption underestimates variability during peak flows.

The normal observational model improves interpretability by explicitly modeling uncertainty. While effective for capturing seasonal trends, it is limited in representing extreme events, highlighting the need for alternative error models when high-flow accuracy is critical.

Lognormal Observation Model



Here we calibrated the HBV model using a lognormal likelihood function, which assumes that residuals follow a lognormal distribution. This is particularly suitable for hydrological variables like streamflow, which are non-negative and often right-skewed.

The resulting plot shows observed discharge (red) along with the 5–95% confidence interval (grey) based on the lognormal distribution of simulated values. Compared to the normal model, the lognormal approach produces wider and more asymmetric uncertainty bands, especially during high-flow events. These bands better accommodate the variability in peak flows, which are systematically underestimated in the normal model.

The optimized log-likelihood value (-1.88) is higher (better) than that of the normal model (-2.78), indicating that the lognormal model provides a statistically superior fit to the observed discharge.

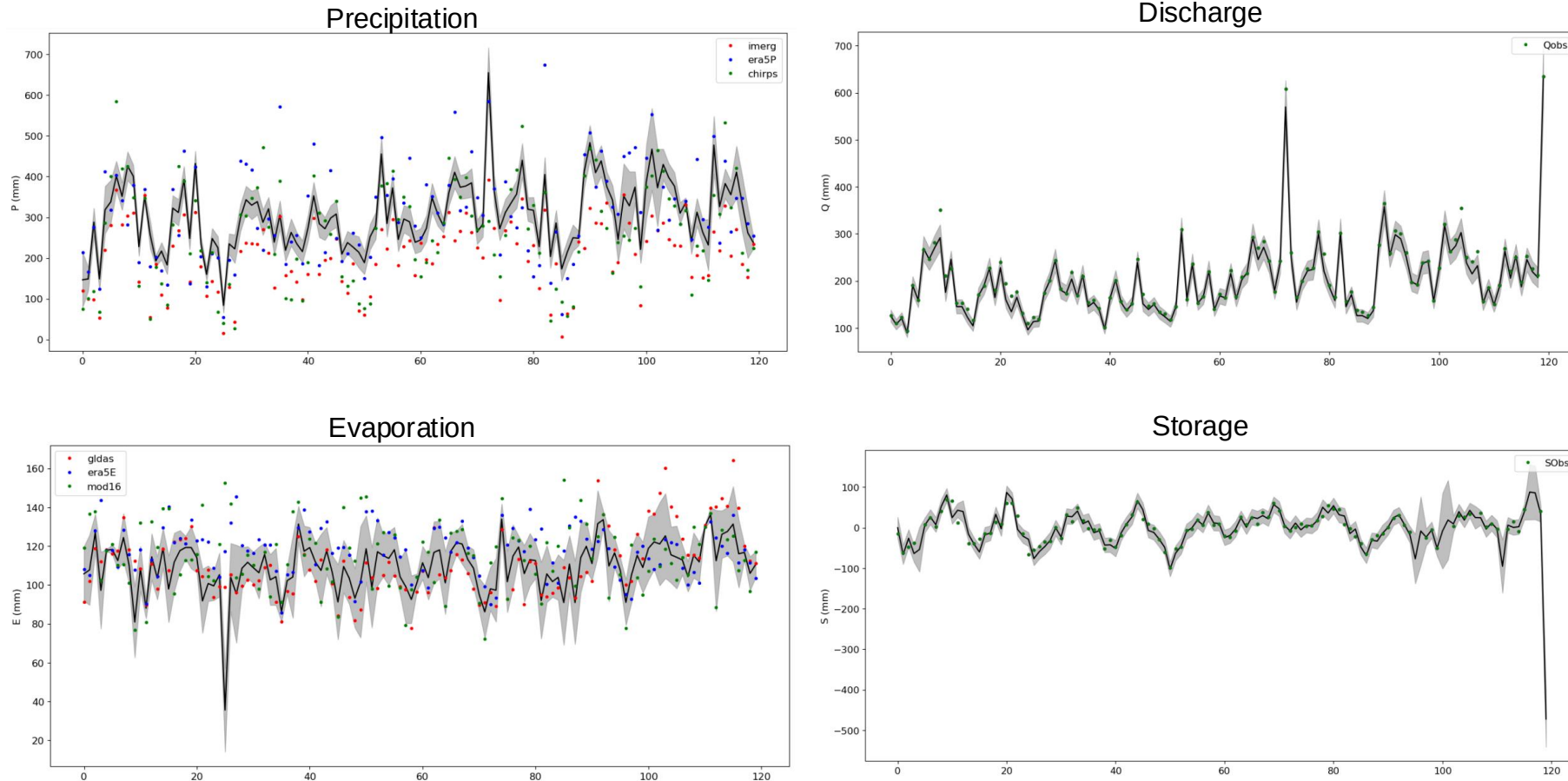
08 Closing Water Balance Using Remote Sensing Data - Data

Data used in this exercise

The data sources used from Earth Engine are:

- Precipitation: 'NASA/GPM_L3/IMERG_V07', 'ECMWF/ERA5_LAND/DAILY_AGGR', 'UCSB-CHG/CHIRPS/DAILY'
- Evaporation: 'NASA/GLDAS/V021/NOAH/G025/T3H', 'ECMWF/ERA5_LAND/DAILY_AGGR', 'MODIS/006/MOD16A2'
- Storage: 'NASA/GRACE/MASS_GRIDS/MASCON_CRI' (lwe_thickness and uncertainty)

The Kalman filtering and smoothing is used to close the water balance. It can be seen that there is quite some variance between the different data sources and especially the precipitation data all falls quite often outside of the post exterior variance band.



09 FlexTopo - Modelling

Data used in this exercise

Cagayan de Oro:

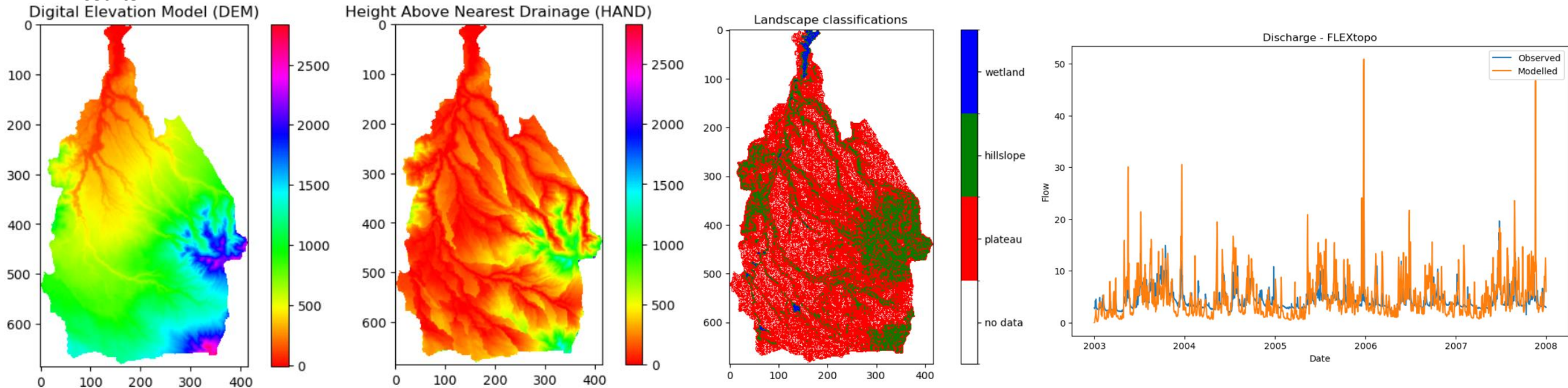
- **Forcing data period:** 2003 - 2008
- **Data type:** precipitation, evaporation (ERA5), discharge (Philippine Government)
- **QGIS files:** hand.asc, slope.asc, basin.asc, facc.asc, fdir.asc, dem.asc

Analysis

Our catchment has a lot higher elevation than the one used in the example exercise. Therefore our hand and landscape classification looks a bit different, but it does comply with the data from QGIS.

In the second part, where the discharge is simulated the observed discharge and ERA5 precipitation and evaporation data are used. It should be noted that the parameters from the exercise are used. Ideally, they should be calibrated for our region, like done for the hbv model. Since FlexTopo will not be used for answering the research question, the decision is made to use our time for the research question instead of calibrating the FlexTopo model. A calibrated model is expected to show a better modelled discharge.

Results



10 Land Use Classification

Data used: Landsat Data (<https://earthexplorer.usgs.gov/>)

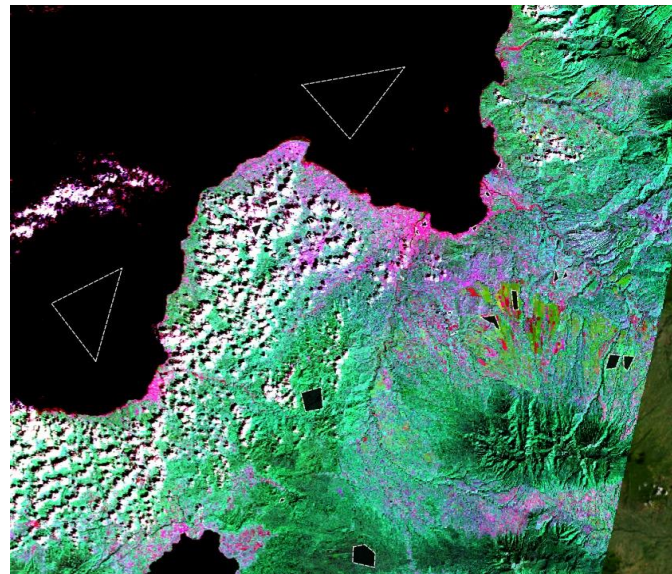
Software used: QGIS

Using bands 1-7 from Landsat data, colours were assigned as followed: Band 5 to Red, Band 6 to Green, and Band 4 to Blue. Next a training set was built, by outlining small test areas for different land classifications (urban, forest, clouds, water and agriculture). Within these classes also several different types were defined. After this the signatures statistics were extracted for the test areas and used to perform a maximum likelihood classification. Finally, the area entire area was classified according to the different land types.

The classification works well for large water and forest areas, and for other large areas of each of the subclasses. However, where the area is more variable with different land types within, especially in the agricultural areas, the classification is not very accurate.

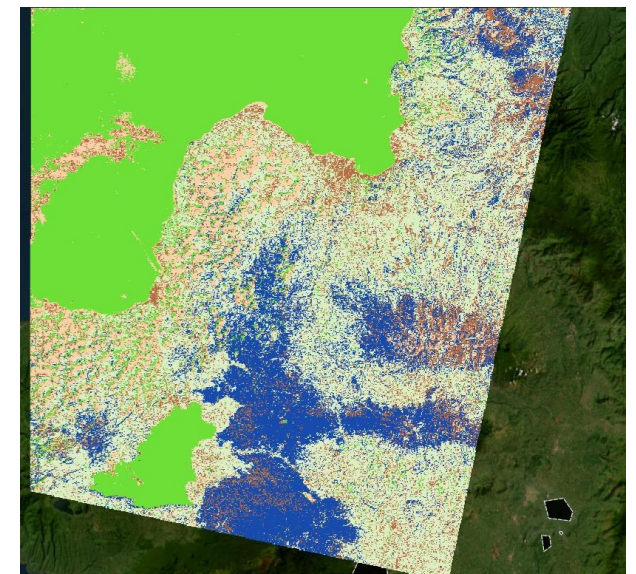


(Area selection and colour composite)



(Test areas)

- ✓  **Classification1555**
- Band 1 (Gray)
-  0 - Unclassified
 -  1 - Water
 -  3 - Urban
 -  2 - Forest
 -  4 - Cloud
 -  5 - Agri



(Classification results)